

Dinghao Luo

Neuroscientist | Machine Learning, Statistical Modelling & Scientific Software
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Open to relocation in Europe · Available January 2027; flexible after October 2026 PhD defence

Neuroscience PhD candidate with six years of experience developing statistical analyses, machine-learning tools and scientific software for behavioural, time-series, and imaging data.

CORE SKILLS

Programming	Python (PyTorch, scikit-learn, statsmodels, pandas), SQL (PostgreSQL), R, MATLAB
Modelling & evaluation	Neural networks, regression, classification, GLMs, Bayesian decoding, time-series analysis, permutation inference, held-out and temporal validation
Software	Qt, Git, automated testing, GitHub Actions, CLI and package development

EXPERIENCE

Graduate Researcher · Max Planck Florida Institute for Neuroscience May 2021–Dec 2026

- Built reproducible Python pipelines processing 50+ TB of behavioural, electrophysiological, and imaging data, including event alignment, feature extraction, time-series analysis, dimensionality reduction and population-level modelling.
- Designed controlled experiments across 100+ animal recording sessions using electrophysiology, optogenetics, pharmacology, two-photon imaging, and virtual reality to test effects on neural activity and behaviour.
- Used GLMs, matching, and permutation inference to test competing mechanisms; rejected the original hypothesis when its predicted behavioural effect did not appear, then open-sourced and documented the analysis workflows for a first-author manuscript submitted to *Cell*.

Undergraduate Researcher · University of Cambridge, Paulsen Lab July 2019–Apr 2020

- Designed electrophysiology experiments on dopamine and synaptic plasticity; First-Class thesis.

Research Intern · UCL Queen Square Institute of Neurology July–Aug 2018

- Quantified biomarker activity in Down syndrome models; co-authored peer-reviewed paper.

SELECTED PROJECTS

FibreSight · github.com/dinghaoluo/fibre-sight

- Built and open-sourced a PyTorch U-Net and Qt application for axon segmentation in microscopy images; achieved 0.93 recall and 0.83 F₂ on nine held-out sessions, reducing curation time from days to hours.
- Packaged labelling, training, prediction, and correction into GUI and CLI workflows with tests, GitHub Actions, and a model card; selected a recall-biased operating point because missed axons (false negatives) required time-consuming manual annotation, whereas false positives could be deleted quickly.

Kairos · github.com/dinghaoluo/kairos

- Built an Python machine learning pipeline using OpenAlex data with reviewed labels, lagged features and expanding-window validation; temporal baselines showed that calendar time explained most of the apparent predictive signal so far.

The Zone · dinghaoluo.github.io/the-zone-site

- Built an Astro/D3 visual companion to Pynchon's *Gravity's Rainbow* using a curated entity registry, deterministic context-aware resolution of character mentions, graph analysis and interactive network and lexical visualisations.

EDUCATION

PhD Candidate, Systems Neuroscience · FAU / Max Planck Florida Institute for Neuroscience 2021–2026

- IMPRS Fellow; defence scheduled for October 2026.

BA Biological Natural Sciences, First-Class Honours · University of Cambridge 2017–2020

- Jardine Scholar (full merit funding); Bundy Scholarship.

OTHER

Translator of *Entangled Life* (Sheldrake, 2020), Chinese edition 《菌络万象》; Pingshan Book Award 2025.
Science writer and former editor at *Scientific American China* (2020–2021) and *Neu-Reality* (2017–2021).
Native Cantonese and Mandarin; fluent English; conversational German.